Project 2: Mechanical & Civil C++ Engineering Project

Project Brief Email

From: design-group@apex-structural.co
To: Structural Computational Analyst
Subject: Project Request: Finite Element Truss Node Stress Solver

Dear Engineering Consultant,

Apex Structural is verifying the structural margin of safety for a new long-span roof truss system. The structural design contains critical nodal coordinates subjected to multi-axis mechanical loads.

We need a structural evaluation program written in modern C++ that processes point vectors, models external vector loading profiles, resolves internal stresses across elements, and outputs structural safety margins.

The software must handle matrix-like collections of forces natively without relying on heavy external runtime frameworks, maintaining a modular design throughout. Please provide the C++ script along with the engineering validation documentation.

Sincerely,
Apex Structural Core Design Team